
SDS210: PROGRAMMING WITH SPATIAL DATA

PROJECT'S REPORT

Live statistics and visualization of wildfires across Europe

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General information

- GitHub repository link: https://github.com/jo1266/Programming_with_spatial_data.git
- Deployed app link: <https://wildfires-across-europe.streamlit.app/>
- Selected spatial problematic: where wildfires are currently happening across Europe, when have they started, and what are their potential consequences on air quality?
- Goal to solve it: fetch live data via a FIRMS API, visualize it on different map kinds, provide a windy map to assess how these fires could evolve, and display an air quality map to grasp potential fire consequences

Workflow and implementation

- Started by running the code example publicly available on the FIRMS website to get wildfires data and visualize it on static maps.
- Adopted an iterative approach to modify it in a more personalized and complex way with targeted help of AI.
- Incrementally added interactive functionalities for users to have a smoother experience.
- Regularly added code comments to help future developers to understand the code.
- Ended up by deploying my Streamlit app as a publicly available website.

Key results

- FIRMS data fetching seamlessly works for the selected country
- Wildfires visualization is split into 3 different maps: a basic one, a categorical one highlighting when these fires started, a multi-layered one allowing terrain and satellite visualization
- Embedded windy interactive map focused on the selected country that allows users to switch between relevant layers.
- Added an air quality map relying on open-meteo live data allowing users to switch between different air quality variables.

Challenges and solutions

- Challenge 1: set up a proper EU clickable map allowing users to effectively click on a specific country
- Solution 1: Iterative trial-and-error with the help of AI and eventually solved by properly inspecting the map data structure allowing to correctly return countries' coordinates
- Challenge 2: Saving some variables so that users can switch between pages without needing to re-select or re-fetch everything
- Solution 2: efficient use of the "st.session_state" function. Note that the selected dataset and time frame on page 1 reinitialize when users come back on page 1. This remain an unsolved bug unfortunately...
- Challenge 3: properly deploy the app as a public website

- Solution 3: added a "requirements.txt" and "packages.txt" file to indicate the required library to fetch.

Use of AI tools

The great majority of the code used within the app has been generated by AI tools. This "raw" code has then been reviewed, tweaked, and adapted by myself. Indeed, I had very specific needs so going through every code lines and modify some parts has been an essential process. I also used AI tools to debug heavy problems faster. Smaller bugs have been solved manually. All the rest of this project is the product of my personal reflection, logic, and work.

References